

SEMESTER 6

Advanced Algorithms

Course Code	HNCST609	CIE Marks	40
Teaching Hours/Week (L:T:P)	3:1:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCST502	Course Type	Theory

Course Objectives:

1. To introduce advanced algorithmic techniques and theoretical tools for analyzing complex problems.
2. To enable students to design and evaluate efficient solutions using dynamic, greedy, randomized, and approximation strategies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Advanced recurrence forms, Akra-Bazzi Theorem, Dynamic Programming - Longest Common Subsequences, Bellman Ford algorithm, Backtracking - Subset Sum, Hamiltonian Path (concept only), Branch and Bound - Knapsack problem, Greedy Method - Huffman codes, Matroids. Space-Bounded Computations - Basic concepts of L & NL, Space hierarchy (introductory), Savitch's Theorem (concept only).	11
2	Sorting Networks - Comparison Networks, Zero-One Principle, Bitonic Sorting Network - structure & analysis, Merging Networks, Batcher's Odd-Even Mergesort Network, Complexity of Sorting Networks (depth & size). String Matching Algorithms - The Naïve Pattern Matching Algorithm, Rabin-Karp Algorithm - analysis using hashing, Finite Automata-based String Matching, Knuth -Morris-Pratt (KMP) Algorithm - prefix function, analysis.	11
3	Randomization - Basic Probability - indicator variables, inequalities & Bounds - Markov's Inequality, Chebyshev's Inequality, Chernoff Bound (applications only) - Universal Hashing - Expectations, Markov Chains and Random Walks, 2-SAT random walk, random walks on graphs (concept only), Applications of Randomized Algorithms.	11

4	Approximation Algorithms - Approximation Algorithms for NP -Hard Problems - Approximation Algorithms for the Vector cover problem, Traveling Salesman Problem - Knapsack Problem, Algorithms for Solving Nonlinear Equations - Bisection Method - Method of False Position - Newton's Method.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain advanced algorithmic concepts such as recurrence solving, sorting networks, randomized bounds and space-bounded computations.	K2
CO2	Summarize key algorithmic strategies including dynamic programming, greedy methods, backtracking and approximation techniques.	K2
CO3	Apply dynamic programming, greedy and randomized methods	K3

	to solve problems such as LCS, Bellman-Ford, KMP and hashing-related tasks.	
CO4	Apply approximation algorithms and numerical methods to obtain solutions for NP-hard problems and nonlinear equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							2
CO2	3	3	3	2							2
CO3	3	3	3	2	2						2
CO4	3	3	3	2							2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein	MIT Press	4/e, 2022
2	Randomized Algorithms	Rajeev Motwani and Prabhakar Raghavan	Cambridge University Press	1/e, 2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Algorithms	Robert Sedgewick and Kevin Wayne	Addison-Wesley	4E, 2011
2	Algorithm Design	Jon Kleinberg & Éva Tardos	Addison-Wesley	1E, 2006
3	Introduction to the Theory of Computation	Michael Sipser	Cengage Learning	3E, 2012
4	Approximation Algorithms	Vijay V. Vazirani	Springer-Verlag Berlin Heidelberg	1E, 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_cs39
2	https://onlinecourses.nptel.ac.in/noc23_cs01
3	https://www.digimat.in/nptel/courses/video/106105225
4	https://onlinecourses.nptel.ac.in/noc24_cs97

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
SIXTH SEMESTER B. TECH HONOURS DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: HNCST609				
Course Name: Advanced Algorithms				
Max. Marks: 60			Duration: 2 hours 30 minutes	
		PART A		
		Answer all questions. Each question carries 3 marks	CO	Mark
1		Differentiate between L and NL with suitable examples.	1	(3)
2		Explain how Bellman–Ford algorithm applies dynamic programming to handle negative edge weights.	3	(3)
3		State the Zero-One Principle and explain its significance in sorting networks.	1	(3)
4		Construct the prefix function (π table) for the pattern “ ababaca ” using the KMP algorithm.	3	(3)
5		State Markov’s Inequality and Chebyshev’s Inequality with their conditions of applicability.	1	(3)
6		Given a hash family H, show how Universal Hashing reduces collision probability in randomized algorithms.	3	(3)
7		Define approximation ratio. Illustrate with an example for a simple NP-hard problem.	2	(3)
8		Apply the Bisection Method to approximate a root of the equation $f(x)=x^3-4x+1$ in the interval $[0, 2]$ for one iteration.	4	(3)

PART B				
<i>Answer any one full question from each module. Each question carries 9 marks</i>				
Module 1				
9	a)	Explain the Akra–Bazzi Theorem. Illustrate with the recurrence: $T(n)=T(n/2)+T(n/3)+n$	1	(4)
	b)	Using backtracking, solve the Subset Sum problem for: $S = \{3, 5, 7, 10\}$, Target = 15. Show the search tree up to the first solution.	3	(5)
10	a)	Describe the structure of matroids. Explain how the greedy method is justified using matroid properties.	1	(4)
	b)	Apply Branch and Bound to solve the 0/1 Knapsack problem for the following items: Weights = {2, 3, 4, 5}, Profits = {3, 4, 5, 8}, Capacity=5. Draw the state-space tree and bound calculations.	3	(5)
Module 2				
11	a)	Explain the structure and working of a Bitonic Sorting Network with neat diagram.	1	(4)
	b)	Construct the DFA table for the Finite Automata-based String Matching algorithm for the pattern “aba” over alphabet {a, b}.	3	(5)
12	a)	Explain Batcher’s Odd-Even Mergesort Network. Discuss its depth and size.	1	(4)
	b)	Given the text $T = \text{“3245532453”}$ and pattern $P = \text{“245”}$, perform Rabin Karp pattern matching with modulo 11. Show the rolling hash calculations.	3	(5)
Module 3				
13	a)	Define indicator variables and derive expected value of the number of heads in ‘n’ biased coin tosses.	1	(4)
	b)	Explain the randomized algorithm for 2-SAT using random walks. Apply one iteration to a small example.	3	(5)
14	a)	Explain Markov Chains and stationary distributions with a simple example.	1	(4)
	b)	Apply Chernoff Bound to estimate the probability that the number of heads in 100 fair coin tosses deviates from the mean by more than 20.	3	(5)
Module 4				

15	a)	Explain the approximation algorithm for the Vertex Cover problem. Discuss approximation ratio.	2	(4)
	b)	Using Newton's Method, compute one iteration to find the root of $f(x) = x^2 - 5x + 6$, starting at $x_0 = 0$.	4	(5)
16	a)	Describe the approximation algorithm for metric TSP using Minimum Spanning Tree.	2	(4)
	b)	Solve one iteration of the Method of False Position for $f(x) = x^3 - x - 1$, over the interval $[1, 2]$.	4	(5)
